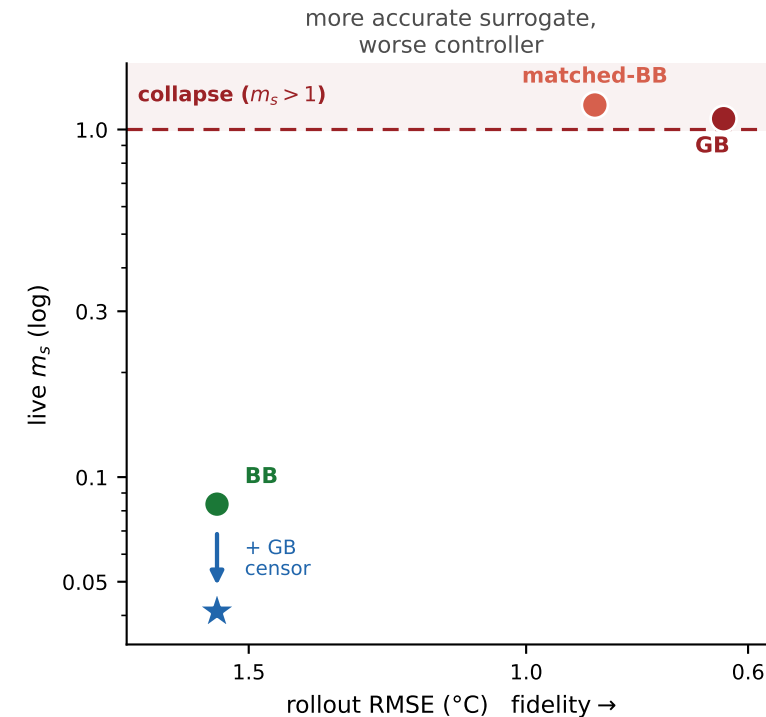
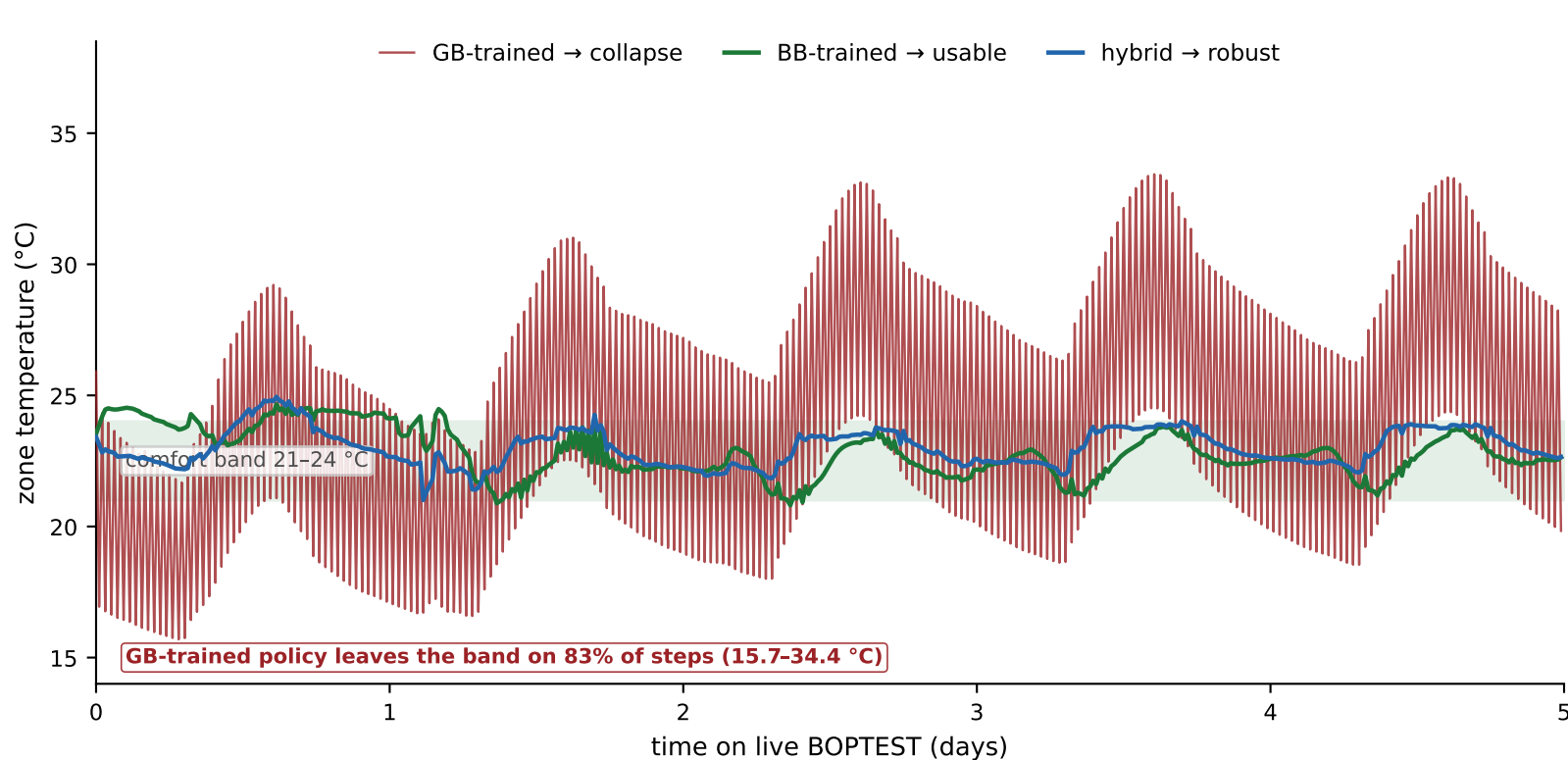


The Fidelity-Utility Paradox in Surrogate-Based RL for HVAC Control

The controller trained on the most accurate surrogate (GB) drives the live zone out of comfort; the coarse surrogate (BB) and the hybrid keep it in band.



Live BOPTEST outcomes from committed closed-loop traces (peak-heat window). BB = coarse black-box (hourly); matched-BB = same architecture at 15 min; GB = calibrated grey-box twin.